

ZAIN SAJID

Software Engineer | React.js • Node.js • MERN • REST APIs • AI Integration

Lahore, Pakistan | 0308-4042598 | zainsajid.056@gmail.com

linkedin.com/in/zain-sajid12 | github.com/zain-sajid-18 | zain-sajid-portfolio.vercel.app

PROFESSIONAL SUMMARY

Results-driven Full Stack MERN Developer and recent Software Engineering graduate from the University of Central Punjab (graduated July 2026), with proven experience architecting, deploying, and maintaining scalable production-grade web applications and REST APIs. Proficient in real-time systems (Socket.io), AI-integrated platforms (OpenAI API), cross-platform mobile development (React Native), and responsive React.js frontends. Passionate about clean modular architecture, database optimization, and delivering measurable business impact through technology.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, C++, Java

Frontend: React.js, Next.js, Tailwind CSS, HTML5, CSS3

Backend: Node.js, Express.js, Nest.js, REST APIs, JWT Auth

Mobile: React Native (cross-platform iOS & Android)

Databases: MongoDB, MySQL, Firebase, Mongoose ODM

Real-Time: Socket.io, WebSockets, Event-Driven Architecture

DevOps & Cloud: Docker, CI/CD Pipelines, AWS, Vercel, Render

AI Integration: OpenAI API, REST integration, MERN deployment

Testing & Tools: Jest, Selenium WebDriver, Git, GitHub, Postman, VS Code, npm, Linux CLI

Concepts: MVC, OOP, Design Patterns, Agile, System Design

KEY PROJECTS

IntelliBid — AI-Powered Online Auction Platform

2025 – 2026

React.js • Node.js • Express.js • MongoDB • OpenAI API • Socket.io

- Architected an end-to-end AI-driven auction platform with a real-time **bidding engine via Socket.io** — managing concurrent bids, bid-expiry countdowns, anti-sniping logic, and live leaderboard synchronization for hundreds of simultaneous users with zero data inconsistency.
- Integrated **OpenAI API** for intelligent product description generation, smart bid-price recommendations, and automated item categorization — significantly improving seller productivity and platform discoverability.
- Built secure **JWT authentication with role-based access control** (Admin / Seller / Buyer), complete item CRUD with multi-image upload, payment-ready checkout flow, and a real-time React.js dashboard with live toast notifications and auction-status tracking.
- Designed optimized **MongoDB schemas** with compound indexing and aggregation pipelines for bid-history analytics — maintaining sub-100ms API response times under peak concurrent bidding load.

Real-Time Chat Application

2026

Node.js • Socket.io • MongoDB • JWT • React.js

- Developed a production-grade messaging platform with persistent **WebSocket connections**, live typing indicators, user presence detection, read-receipts, file sharing, and fully paginated chat history stored in MongoDB.
- Implemented **JWT-secured REST endpoints**, bcrypt password hashing, and XSS-safe rendering — achieving sub-50ms real-time message delivery latency under concurrent multi-room load.

E-Commerce Platform

2024

React.js • Node.js • Express.js • MongoDB • Stripe-ready

- Delivered a full-featured e-commerce system with product catalogue, advanced search & filtering, cart and wishlist management, order lifecycle tracking, and a **role-based admin analytics dashboard** — cutting average API response time by 40% through compound indexing and query optimization.

Scalable REST API System

2024

Node.js • Express.js • MongoDB • JWT • Modular Architecture

- Engineered a production-ready modular RESTful API with **layered architecture** (routes → controllers → services → models), Joi input validation, centralized error handling, JWT middleware, rate-limiting, and comprehensive Postman collection documentation for team onboarding.

EDUCATION

BS Software Engineering — University of Central Punjab (UCP), Lahore

2022 – Jul 2026

CGPA: 3.28 / 4.0 | Relevant Coursework: Data Structures & Algorithms • OOP • Database Systems • Computer Networks • Web Technologies • Software Engineering